

Assignment No 6

Aim: Implement an interface in Java and create multiple classes that implement the interface, demonstrating polymorphism.

Code:

```
interface Employee_Dept {
    void work();
    double calculateSalary();
    String getDepartment();
}

class ITEmployee implements Employee_Dept {
    private String name;
    private double monthlySalary;

    public ITEmployee(String name, double monthlySalary) {
        this.name = name;
        this.monthlySalary = monthlySalary;
    }

    @Override
    public void work() {
        System.out.println(name + " is writing code and fixing bugs.");
    }

    @Override
    public double calculateSalary() {
        return monthlySalary;
    }

    @Override
    public String getDepartment() {
        return "IT Department";
    }
}

class HREmployee implements Employee_Dept {
    private String name;
    private double monthlySalary;

    public HREmployee(String name, double monthlySalary) {
        this.name = name;
        this.monthlySalary = monthlySalary;
    }
}
```

```
@Override
public void work() {
    System.out.println(name + " is recruiting and managing employees.");
}
```

```
@Override
public double calculateSalary() {
    return monthlySalary;
}
```

```
@Override
public String getDepartment() {
    return "HR Department";
}
}
```

```
class FinanceEmployee implements Employee_Dept {
    private String name;
    private double monthlySalary;

    public FinanceEmployee(String name, double monthlySalary) {
        this.name = name;
        this.monthlySalary = monthlySalary;
    }

    @Override
    public void work() {
        System.out.println(name + " is preparing financial reports and budgets.");
    }

    @Override
    public double calculateSalary() {
        return monthlySalary;
    }

    @Override
    public String getDepartment() {
        return "Finance Department";
    }
}
```

```
public class InterfaceWithPolymorphism {

    public static void main(String[] args) {

        Employee_Dept emp1 = new ITEmployee("Siddharth", 60000);
        Employee_Dept emp2 = new HREmployee("Tejas", 50000);
        Employee_Dept emp3 = new FinanceEmployee("Om", 55000);
```

```
Employee_Dept[] employees = {emp1, emp2, emp3};

for (Employee_Dept emp : employees) {
    emp.work();
    System.out.println("Department: " + emp.getDepartment());
    System.out.println("Salary: $" + emp.calculateSalary());
    System.out.println("-----");
}
}
```

Output:

```
Siddharth is writing code and fixing bugs.
Department: IT Department
Salary: $60000.0
-----
Tejas is recruiting and managing employees.
Department: HR Department
Salary: $50000.0
-----
Om is preparing financial reports and budgets.
Department: Finance Department
Salary: $55000.0
-----

Process finished with exit code 0
```